

Technical Assessment

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Preliminary setup

In the `stevedata` package in R, we load in the wage data from the General Social Survey (1974-2018), illustrating wage gap by gender. (See [here](#) for more detailed documentation.) After we load in the data, we display some summary statistics to begin with.

```
# some helpful libraries
library(stevedata)
library(dplyr)
library(ggplot2)
library(reshape2)
```

```
# load in the data & display summary statistics
data(gss_wages)
summary(gss_wages)
```

year	realrinc	age	occ10
Min. :1974	Min. : 227	Min. :18.00	Min. : 10
1st Qu.:1985	1st Qu.: 8156	1st Qu.:32.00	1st Qu.:2710
Median :1996	Median : 16563	Median :44.00	Median :4720
Mean :1996	Mean : 22326	Mean :46.18	Mean :4696
3rd Qu.:2006	3rd Qu.: 27171	3rd Qu.:59.00	3rd Qu.:6230
Max. :2018	Max. :480144	Max. :89.00	Max. :9997
	NA's :23810	NA's :219	NA's :3561
occrcode	prestg10	childs	wrkstat
Length:61697	Min. :16.00	Min. :0.000	Length:61697
Class :character	1st Qu.:33.00	1st Qu.:0.000	Class :character
Mode :character	Median :42.00	Median :2.000	Mode :character
	Mean :43.06	Mean :1.923	
	3rd Qu.:50.00	3rd Qu.:3.000	
	Max. :80.00	Max. :8.000	

	NA's :4186	NA's :189
gender	educat	maritalcat
Length:61697	Length:61697	Length:61697
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

From these initial summary statistics, we see that there are a lot of missing values, so we subset our data to only include complete observations, which make up 60.27% of the original dataset.

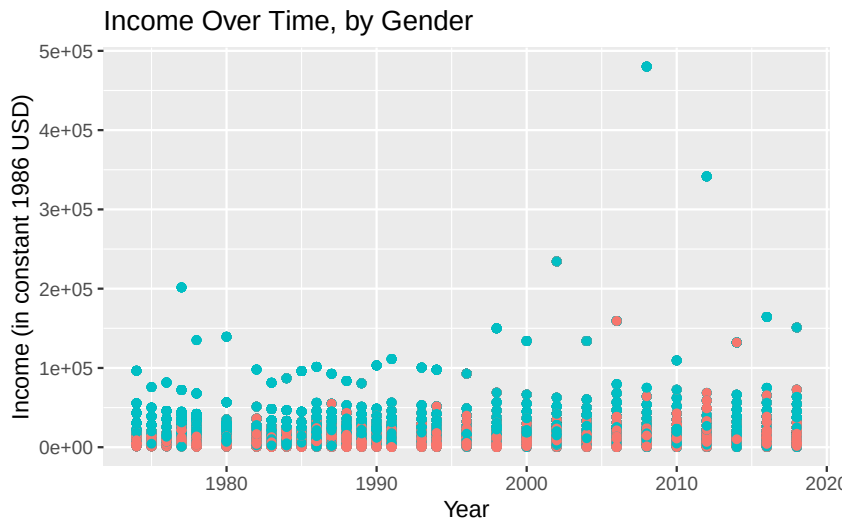
```
# filter to only include complete observations
gss_wages_complete <- gss_wages[complete.cases(gss_wages),]
paste0("Percentage complete observations: ",
       round(100*(nrow(gss_wages_complete))/nrow(gss_wages), 2), "%")
```

```
[1] "Percentage complete observations: 60.27%"
```

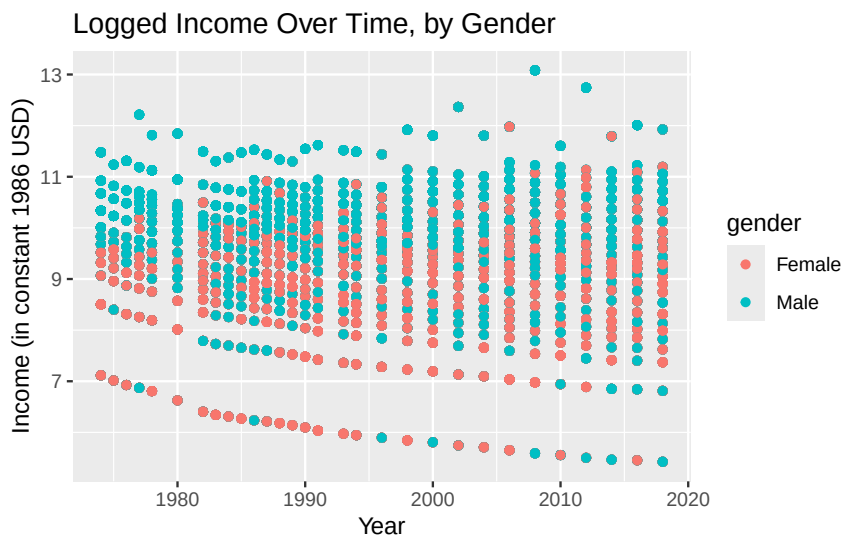
Fact 1: Females have lower income than males.

From the data documentation, we see that it aims to illuminate the wage gap between gender. We start investigating this suggestion by graphing income levels by gender over time:

```
# plot of year v.s. income
ggplot(gss_wages_complete) +
  geom_point(aes(x = year, y = realrinc, color = gender)) +
  labs(x = "Year", y = "Income (in constant 1986 USD)",
       title = "Income Over Time, by Gender") +
  theme(legend.position = "none")
```



```
# plot of year v.s. logged income
ggplot(gss_wages_complete) +
  geom_point(aes(x = year, y = log(realrinc), color = gender)) +
  labs(x = "Year", y = "Income (in constant 1986 USD)",
       title = "Logged Income Over Time, by Gender")
```



The initial plot seems to suggest that males (blue) have higher income than females (red), demonstrated by all the top-most dots being the color blue. To take care of outliers and get a better visual of the pattern, we can also plot the logged income values; this graph shows with a little more clarity the trend again of males making more than females.

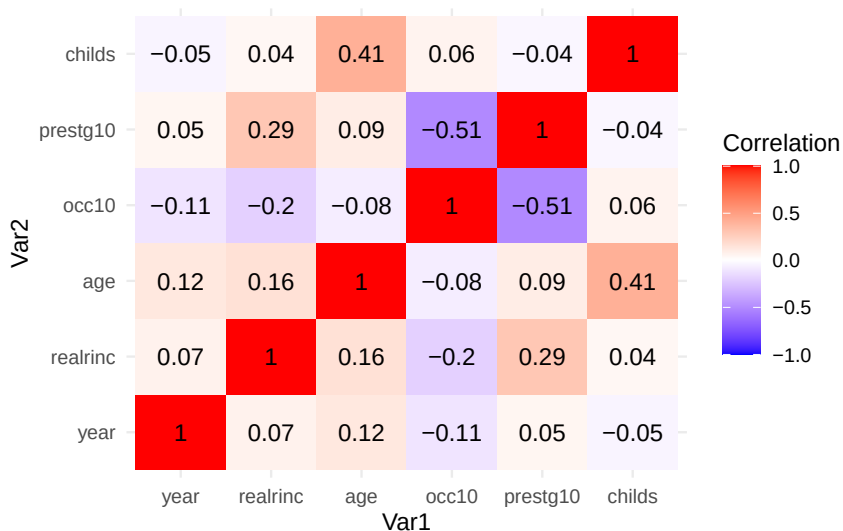
This is important because gender pay equality is something that is supposed to be getting better over time, but this graph suggests that there is still a marked difference.

Fact 2: The number of children has a very small correlation with base income.

From here, one might wonder if any other variables have correlations with each other, or with income in particular. To check this out, we plot a heatmap of correlation values:

```
# correlation matrix
corrs <- gss_wages_complete |>
  select(c(year, realrinc, age, occ10, prestg10, childs)) |>
  cor()

# from matrix, plot heatmap
corrs |>
  melt() |>
  ggplot(aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  geom_text(aes(label = round(value, 2))) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
    limit = c(-1, 1), name = "Correlation") +
  theme_minimal()
```



While one might expect that parents suffer in their careers and therefore income levels to take care of children, we see in particular that **childs** only has a correlation of 0.04 with **realrinc**. Because of this counterintuitive idea, one might hypothesize further that perhaps only *young* children negatively impact income, a distinction that this plot is unable to capture. Additional investigation would be needed. The other variables do not have particularly high correlations with one another either, save for prestige score and occupation, which makes sense (people may see higher paying jobs as more prestigious.)

This is important to investigate because a drop in income due to having children may decentivize individuals trying to raise families, a very pertinent issue in the light of dropping birthrates.